

◦ Aim:

- To determine the value of Planck's constant (h), and to find the values of threshold frequency (ν_0) and the work function (ϕ_0).

◦ Apparatus:

- Photocell, mercury vapour lamp, DC voltage source, digital voltmeter, a set of filters of different wavelengths, connecting wires, etc.

◦ Theory:

- The photoelectric effect was first observed by Becquerel in 1839. Later, Hertz and Lenard published the first thorough investigations.

- In 1901, Max Planck stated that the energy of a photon is directly proportional to the frequency.

$$\therefore E \propto \nu \Rightarrow E = h\nu$$

He defined the proportionality constant ' h ' and named it 'Planck's constant', which is equal to 6.626×10^{-34} Js.

- In 1905, Einstein gave the following expression for the photoelectric:

$$h\nu = h\nu_0 + \frac{1}{2}mV_{\max}^2$$

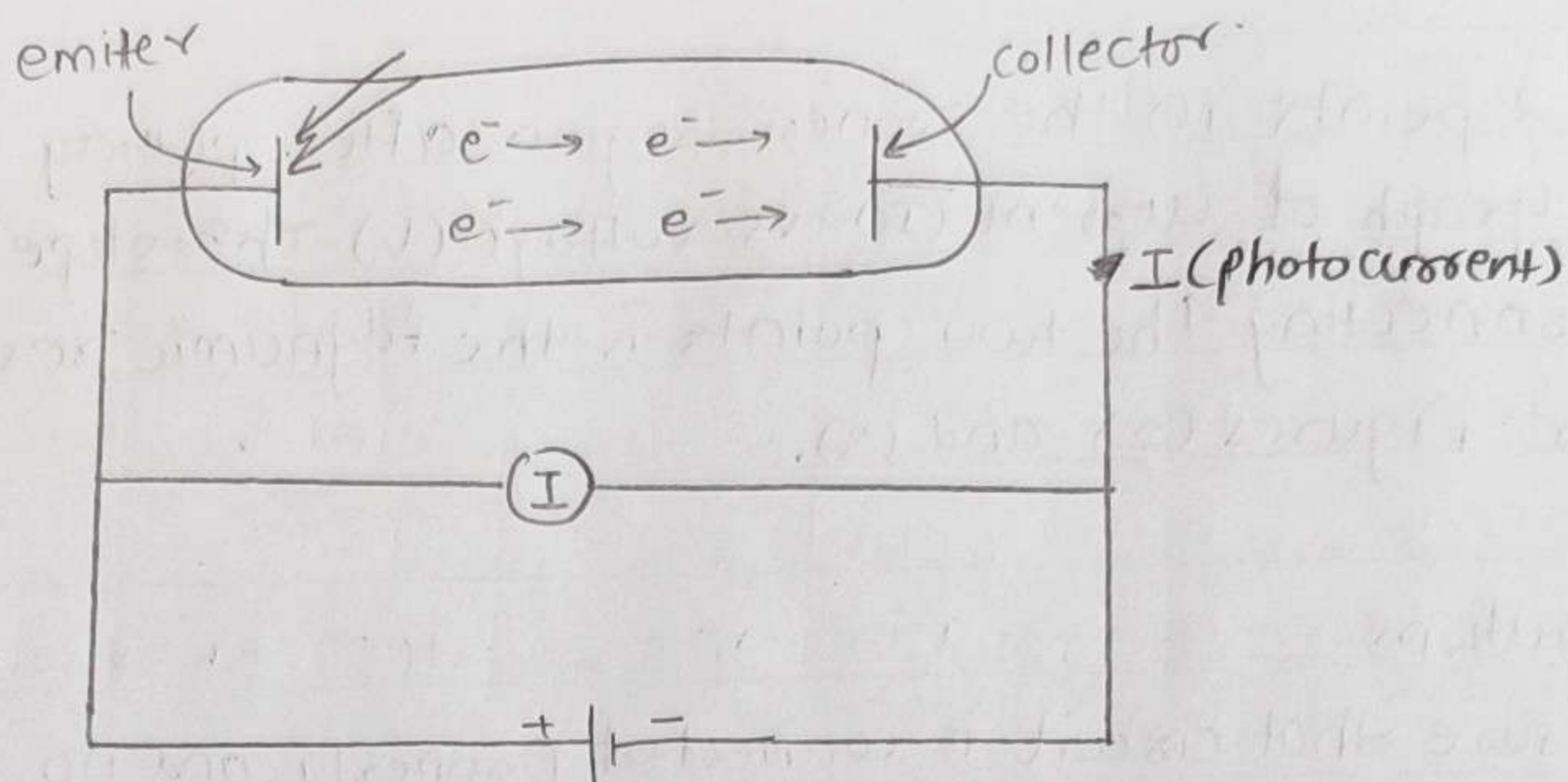
where $V_{\max}$ is the max. velocity of emission of photoelectron.

ν is frequency of the incident light

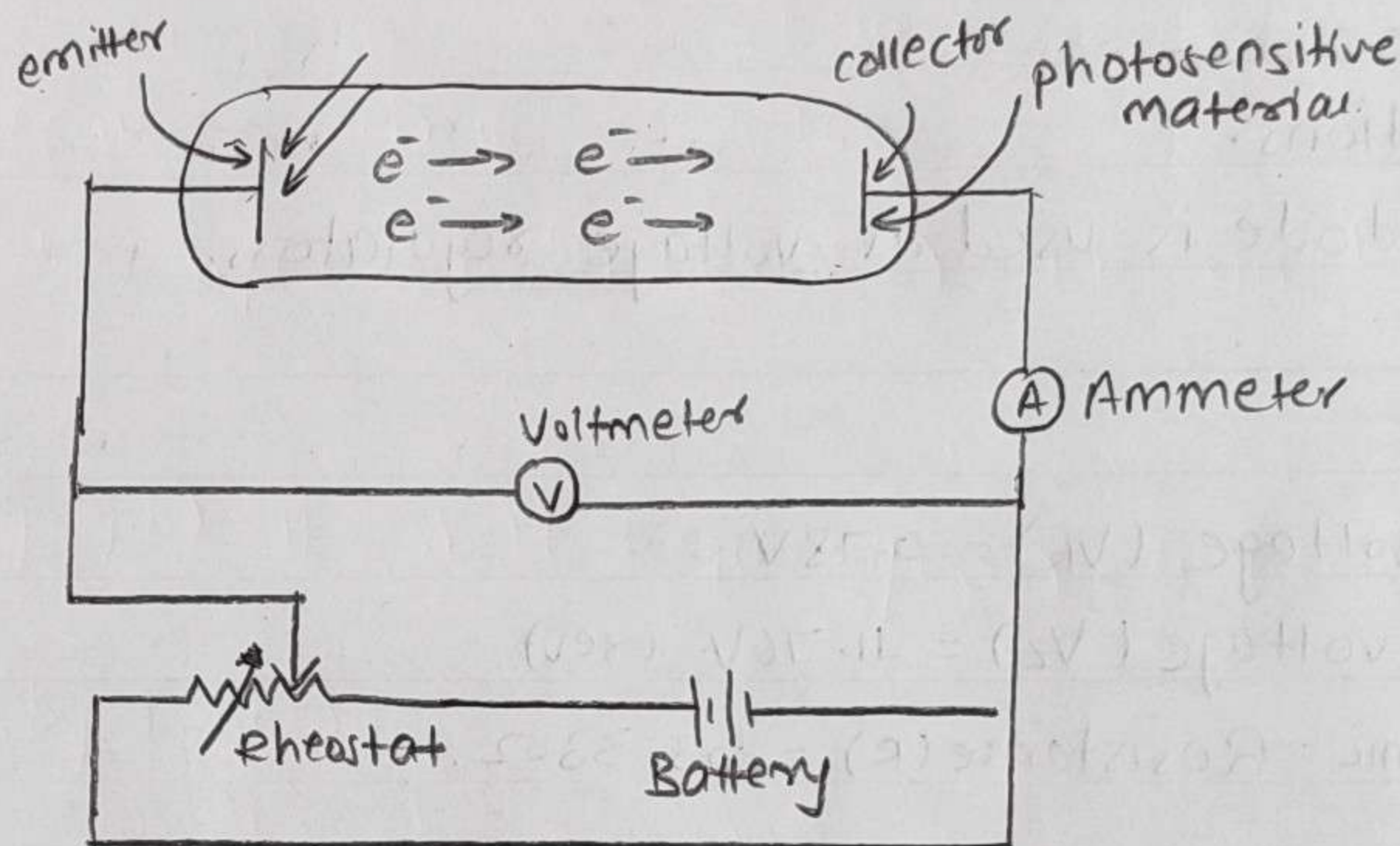
ν_0 is the 'threshold frequency'.

- when light wave of sufficient energy is incident onto a metal surface, electrons are ejected from it. This is called the 'photoelectric effect'. The ejected electrons are called 'photoelectrons' and the

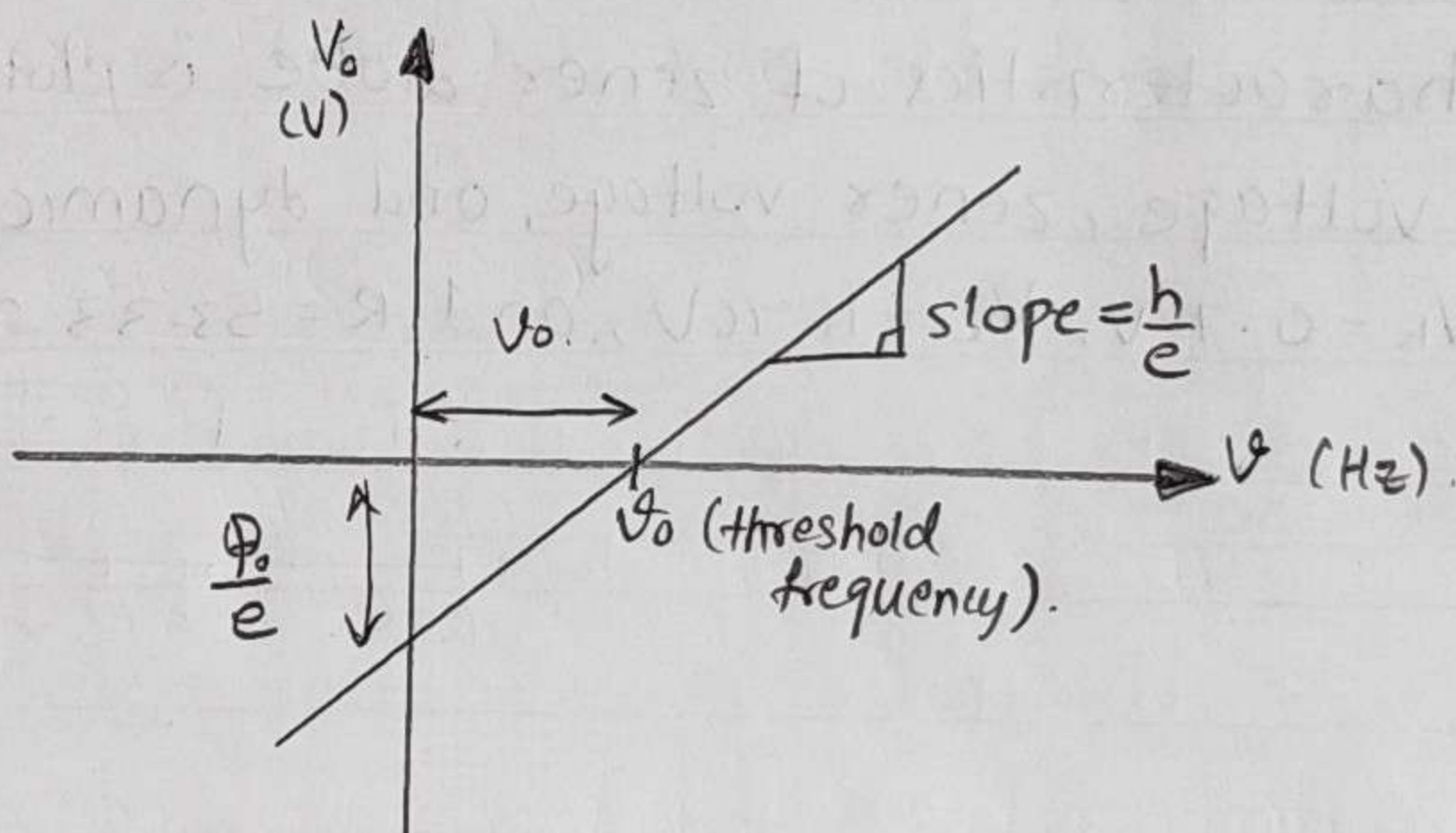
Diagrams:



[Figure (1)]



[Figure (2)]



[Figure (3) - Model Graph]

Observation Table:

Sr. No.	Wavelength ' λ ' (nm)	Frequency ' ν ' (10^{14} Hz)	Stopping Potential		Average Potential $V_0 = \frac{1}{2}(V_{01} + V_{02})$
			V_{01} (V)	V_{02} (V)	
1	490	6.122	0.63	0.65	$\rightarrow 0.64$
2	520	5.769	0.45	0.45	0.45
3	545	5.504	0.37	0.38	0.375
4	560	5.357	0.30	0.30	$\rightarrow 0.30$

Calculations:

$$(i) \text{ slope (by graph) } = \frac{0.64 - 0.30}{(6.122 - 5.357) \times 10^{14}} = \frac{0.34 \times 10^{-14}}{0.765} = 4.44 \times 10^{-14}$$

$$\therefore \frac{h\nu}{e} = 4.44 \times 10^{-14} \Rightarrow h\nu = (4.44 \times 10^{-14})(1.6 \times 10^{-19}) = 7.104 \times 10^{-34} \text{ Js.}$$

$$(ii) \text{ x-intercept (by graph) = threshold frequency } (\nu_0) = 4.75 \times 10^{14} \text{ Hz.}$$

(iii) Calculating 'h' (by table):

$$i) h_1 = \frac{eV_1}{\nu_1 - \nu_0} = \frac{(1.6 \times 10^{-19})(0.64)}{(6.122 - 4.75) \times 10^{14}} = 7.463 \times 10^{-34} \text{ Js.}$$

$$ii) h_2 = \frac{eV_2}{\nu_2 - \nu_0} = \frac{(1.6 \times 10^{-19})(0.45)}{(5.769 - 4.75) \times 10^{14}} = 7.065 \times 10^{-34} \text{ Js}$$

$$iii) h_3 = \frac{eV_3}{\nu_3 - \nu_0} = \frac{(1.6 \times 10^{-19})(0.375)}{(5.504 - 4.75) \times 10^{14}} = 7.257 \times 10^{-34} \text{ Js}$$

$$iv) h_4 = \frac{eV_4}{\nu_4 - \nu_0} = \frac{(1.6 \times 10^{-19})(0.30)}{(5.357 - 4.75) \times 10^{14}} = 6.099 \times 10^{-34} \text{ Js}$$

$$\therefore h_{avg} = \frac{h_1 + h_2 + h_3 + h_4}{4} = \frac{(7.463 + 7.065 + 7.257 + 6.099) \times 10^{-34}}{4} = \frac{27.884 \times 10^{-34}}{4}$$

$$\therefore h_{avg} = 6.971 \times 10^{-34} \text{ Js.}$$

produced current is called the 'photoelectric current'.

- The minimum amount of energy required to eject an electron from the surface of a metal is called the 'work function'. ($\Phi_0 = h\nu_0$)

- The incident energy is used in two ways:

i) absorbed by electron to eject from metal surface.

ii) remaining energy is used to give KE to the electron.

$$\therefore E = \Phi_0 + KE \Rightarrow h\nu = h\nu_0 + \frac{1}{2}mV_{\max}^2$$

$$\text{Hence, } \frac{1}{2}mV_{\max}^2 = h(\nu - \nu_0).$$

- As the intensity of light increases, more photoelectrons are ejected, which results in higher photoelectric current.

- The KE of electrons only depend upon frequency of light.

- The positive side is connected to the emitter and the negative side is connected to the absorber (Figure 1).

- As negative voltage is increased, the current decreases and at one point it becomes zero. This voltage is called 'stopping potential (V_0)'.

$$\therefore KE = eV_0 \quad (\text{where } e = \text{charge of electron} = 1.6 \times 10^{-19}).$$

$$\therefore h(\nu - \nu_0) = eV_0 \Rightarrow \boxed{h = \frac{eV_0}{\nu - \nu_0}}$$

o Procedure:

- Connect the apparatus as per the circuit shown in figure 2.

$$(iv) h = \frac{h_g + h_{avg}}{2} = \frac{7.104 + 6.971}{2} \times 10^{-34} = 7.0375 \times 10^{-34} \text{ Js.}$$

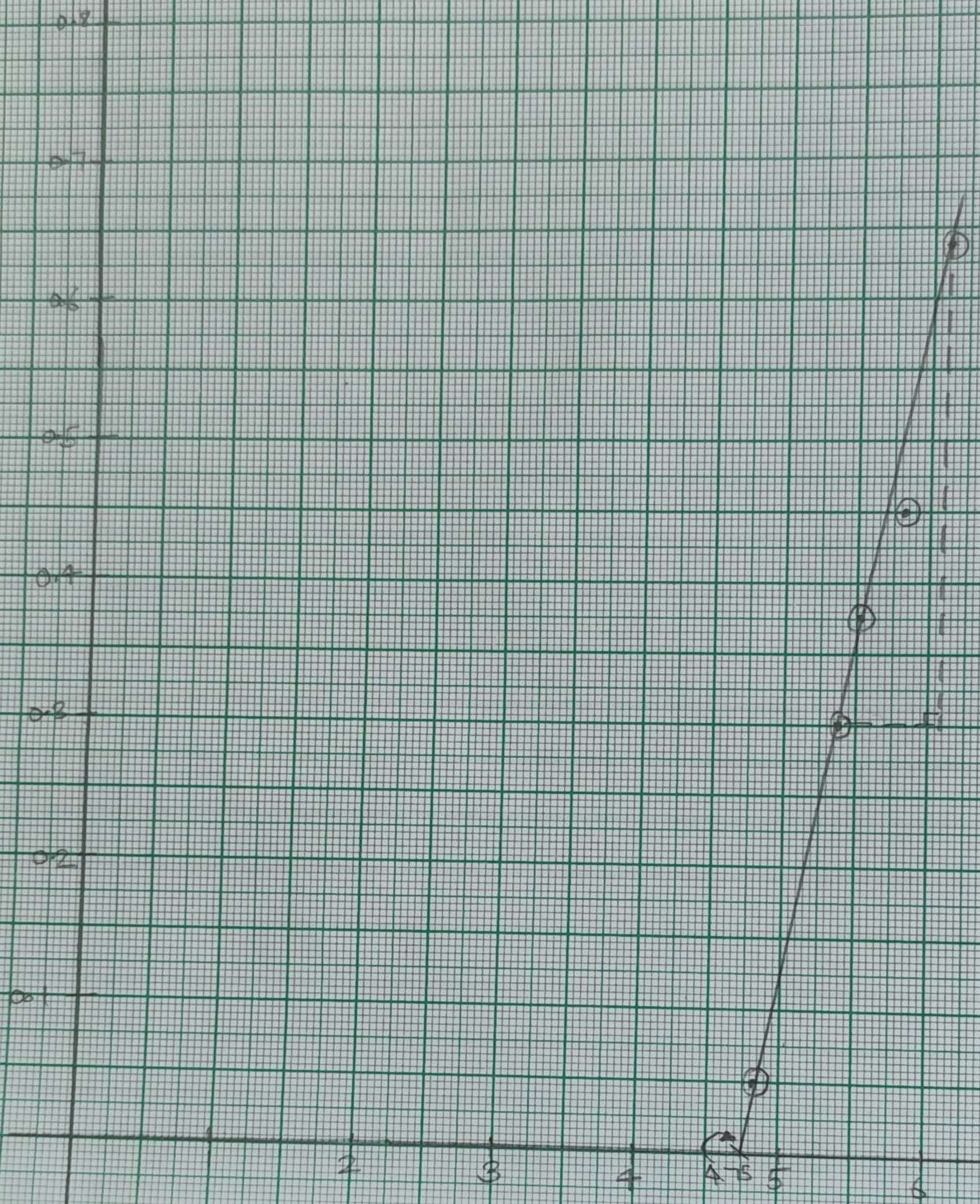
$$(v) \text{ \% error} = \left| \frac{h_{th} - h}{h_{th}} \right| \times 100 = \left| \frac{6.626 - 7.0375}{6.626} \right| \times 100 = 6.210\%$$

$$(vi) \text{ work function, } \phi_0 = \frac{h\nu_0}{e} = \frac{(7.0375 \times 10^{-34})(4.75 \times 10^{14})}{1.6 \times 10^{-19}} = 2.084 \text{ eV.}$$

- Arrange the filters in either ascending or descending order of wavelengths. Select the first filter and insert it into the phototube.
- Increase / Decrease the voltage until the ammeter shows zero current. This voltage is the stopping potential (V_0). Take two trials and the average of the two.
- Repeat the same for the rest of the filters.
- Plot the graph of stopping potential (V_0) against frequency of incident radiation (ν) to get the planck's constant (slope $\times e$) and the threshold frequency (x -intercept).
- Use $h = \frac{eV_0}{\nu - \nu_0}$ to find 'h' values for all the filters and take the average value.
- Find the work function ($\phi_0 = h\nu_0$) and also the % error in the planck's constant: $\left| \frac{h_{th} - h_{calc}}{h_{th}} \right| \times 100$ ($h_{th} = 6.626 \times 10^{-34} \text{ Js}$).
- Precautions:
 - Connect the circuit properly and avoid short circuiting.
 - Do not touch live wires.

Stopping
Potential
 V_0 (V)
↑

Scale: on X-axis, $1\text{cm} = 0.5 \times 10^{14} \text{ Hz}$
on Y-axis, $1\text{cm} = 0.05 \text{ V}$



Incident beam
→

Applications:

- Photoelectric effect is used in nuclear technology.
- It is also used in imaging techniques.

Result:

- Planck's constant (h) = 7.0375×10^{-34} Js.
- Threshold frequency (ν_0) = 4.75×10^{14} Hz
- work function (ϕ_0) = 2.084 eV.

Conclusion:

- Photoelectric Effect is observed.
- Planck's constant (h) is found out to be 7.0375×10^{-34} Js, with an error of 6.210%. from the actual value of 6.626×10^{-34} Js.
- Also, the threshold frequency and work function are found to be $\nu_0 = 4.75 \times 10^{14}$ Hz and $\phi_0 = 2.084$ eV respectively.